

3.6 Torque

Consider a particle of mass m moving about an axis in a circular path of radius r . Let an external force F act on the particle along the tangent to the circular path. The moment of the force $= Fr$. This moment of force is also called torque represented by the symbol τ .

$$\tau = Fr \quad \dots(i)$$

But

$$F = ma = mra$$

$$\tau = (mra)r = mr^2a$$

But

$$I = mr^2$$

$$\tau = Ia \quad \dots(ii)$$

Hence torque is equal to the product of moment of inertia and angular acceleration. Torque is also defined as the rate of change of angular momentum.

$$\tau = \frac{dL}{dt} \quad \dots(iii)$$

$$\tau = \frac{d(I\omega)}{dt} = I \left(\frac{d\omega}{dt} \right)$$

$$\tau = Ia \quad \dots(iv)$$

Hence torque plays the same part in rotatory motion as force in translatory motion. It is a vector quantity. Its dimensions are $[ML^2T^{-2}]$. Its units are $\text{kg-m}^2/\text{s}^2$.

Example 3.1 A. A 0.5 kg. mass is whirled in a circle at the end of a string 0.4 m long, the other end of which is held in the hand. If the mass makes 40 revolutions per second, what is its angular momentum? If the number of revolution decreases by one in 20 s, calculate the mean value of the torque on the system. [B. Sc. (Hons.) 1980]

Here

$$M = 0.5 \text{ kg. } r = 0.4 \text{ m}$$

$$n = 50$$

$$I = Mr^2 \text{ and } \omega = 2\pi n$$

$$\therefore \text{Angular Momentum} = I\omega = Mr^2 \times 2\pi n$$

$$= 2\pi Mnr^2 = 2 \times 3.14 \times 0.5 \times 50 (0.4)^2$$

$$= 25.12 \text{ kg-m}^2/\text{s}$$

Also

$$\omega_1 = 2\pi n_1 \text{ and } \omega_2 = 2\pi n_2$$

$$t = 20 \text{ s}$$

$$\alpha = \frac{\omega_1 - \omega_2}{t} = \frac{2\pi n_1 - 2\pi n_2}{20}$$

$$\alpha = \frac{2\pi (n_1 - n_2)}{20} = \frac{2\pi \times 1}{20}$$

$$= 0.314 \text{ radian/s}^2$$

$$\text{Torque} = Ia = Mr^2 \alpha$$

$$= (0.5)(0.4)^2 \times 0.314$$

$$= 2.512 \times 10^{-2} \text{ kg-m}^2/\text{s}^2$$

3.7A Torque as a Cross Product of $\mathbf{F}$ and $\mathbf{r}$

If a force $\mathbf{F}$ acts an angle θ to $\mathbf{r}$, the torque,

$$\boldsymbol{\tau} = \mathbf{F} \times \mathbf{r}$$

The torque is perpendicular to the plane containing $\mathbf{F}$ and $\mathbf{r}$.

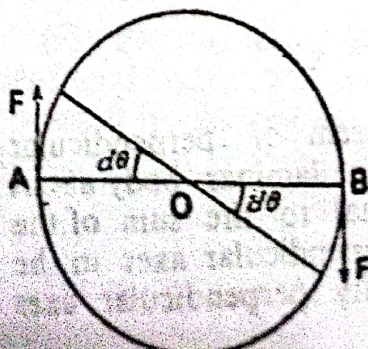
3.7 Analogy Between Translatory Motion and Rotatory Motion

It is noticed that there is an analogy between various physical quantities in translatory motion and rotatory motion. Moment of Inertia in rotatory motion plays the same role as mass in translatory motion. The following table gives the analogy between the various physical quantities in the two types of motion.

S.No.	Translatory Motion	Rotatory Motion
1.	mass m	Moment of Inertia, I
2.	Displacement S	Angular displacement θ
3.	Velocity v	Angular velocity ω
4.	Acceleration a	Angular acceleration α
5.	Force F $F = ma$	Torque τ $\tau = I\alpha$
6.	Linear momentum $p = mv$	Angular momentum $L = I\omega$
7.	Kinetic energy $= \frac{1}{2}mv^2$	Kinetic energy $= \frac{1}{2}I\omega^2$
8.	Work done $= FS$	Work done $= \tau\theta$

3.8 Work Done by a Torque

Consider two particles A and B on the circumference of a disc at diametrically opposite points (Fig. 3.4). Let F and $-F$ be the two forces acting on the particles A and B respectively. If the particles are displaced through $d\theta$, the displacement $dS = r d\theta$ in each case.



Work done by each force

$$= F ds$$

$$= Fr d\theta$$

$$\text{Total work done} = 2 Fr d\theta$$

But $\therefore$ Work done
 If the particles get displaced through an angle $d\theta$ under the action of external torque τ , the total work done
 $= \tau d\theta$... (ii)

When a torque τ acts for a time dt and displaces the particles through $d\theta$,

Work done

$$= \tau d\theta$$

Rate of doing work

$$= \tau \frac{d\theta}{dt} = \tau \omega$$

... (iii)

$$= \text{Torque} \times \text{angular velocity}$$

Example 3.1. A grinder is in the form of a circular disc of mass 10 kg and diameter 0.4 m. Calculate the constant torque that has to be applied so that the disc acquires an angular velocity of 4 revolutions/second in 5 seconds. Calculate also the rate at which work is done by the torque at the end of 5 seconds.

Here $\alpha = \frac{\omega_2 - \omega_1}{t}$

Given $\omega_2 - \omega_1 = 4 \times 2\pi \text{ radians/s}, R = 0.2 \text{ m}$
 $t = 5 \text{ seconds}$

$\therefore \alpha = \frac{8\pi}{5} \text{ radians/s}^2$

$$\text{Torque} = I\alpha = \left(\frac{MR^2}{2}\right) \alpha = \frac{MR^2 \alpha}{2}$$

$$= \frac{10 \times (0.2)^2 \times 8\pi}{5 \times 2} = 1.006 \text{ N-m}$$

Rate at which work is done

$$= \text{Torque} \times \text{angular velocity}$$

$$= (1.006) 8\pi$$

$$= 25.28 \text{ watts}$$

3.9. Theorem of Perpendicular Axes

(a) For a Plane Lamina. The theorem of perpendicular axes states that the moment of inertia of a plane lamina about an axis perpendicular to the plane is equal to the sum of the moments of inertia about two mutually perpendicular axes in the plane of the lamina such that the three mutually perpendicular axes intersect at a common point of intersection.

Mass of the element $= \left(\frac{M}{l}\right) dx$

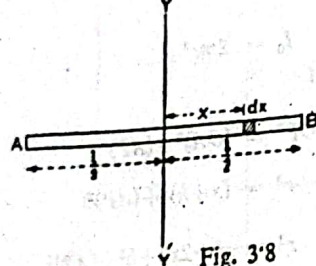


Fig. 3.8

Moment of inertia of the element about the axis YY'

$$= \left[\left(\frac{M}{l}\right) dx \right] x^2$$

Moment of inertia of the bar AB about the axis YY'

$$I = 2 \int_0^{l/2} \left(\frac{M}{l}\right) x^2 dx = \frac{2M}{l} \left[\frac{x^3}{3} \right]_0^{l/2}$$

$$I = \frac{Ml^3}{12} \quad \dots(i)$$

Also

$$I = MK^2$$

$$\therefore MK^2 = \frac{Ml^3}{12}$$

or

$$K = \frac{l}{2\sqrt{3}} \quad \dots(ii)$$

3.12 Moment of Inertia of a Bar AB about an Axis passing through one end and perpendicular to its length (Fig. 3.9).

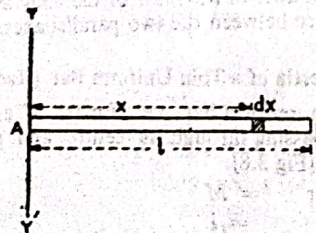


Fig. 3.9

Here

$$I = \int_0^l \left(\frac{M}{l}\right) x^2 dx$$

first

$$I = \frac{Ml^3}{3} \quad \dots(iii)$$

But

$$I = MK^2$$

$$\therefore MK^2 = \frac{Ml^3}{3}$$

$$K = \frac{l}{\sqrt{3}} \quad \dots(iv)$$

3.13 Moment of Inertia of a Bar about an axis perpendicular to its Length, at a distance a from one end (Fig. 3.10)

$$\text{Here } I = \int_{-a}^{(l-a)} \left(\frac{M}{l}\right) x^2 dx$$

$$I = \frac{M}{l} \left[\frac{x^3}{3} \right]_{-a}^{l-a}$$

$$I = M \left[\frac{2}{3} - la + a^2 \right] \quad \dots(v)$$

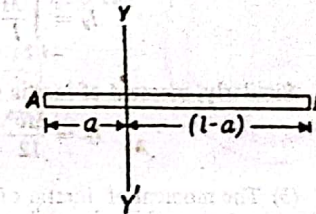


Fig. 3.10

3.14 Moment of Inertia of a Rectangular Lamina

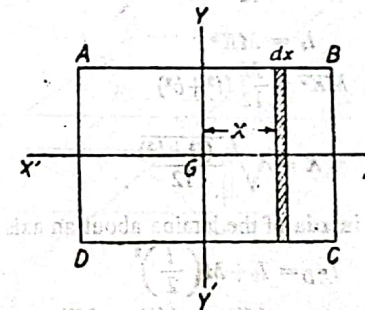


Fig. 3.11

(a) Consider a rectangular lamina of uniform thickness, length l and breadth b . Consider an element of length dx at a distance x from the axis YY' (Fig. 3.11).

$$\text{Mass of the lamina} = M$$

$$\text{Area of the lamina} = l \times b$$

$$\text{Mass per unit area} = \left(\frac{M}{l \times b}\right)$$

$$\text{Area of the element} = b \times dx$$

$$\text{Mass of the element} = \left(\frac{M}{l \times b}\right) b \times dx$$

$$= \left(\frac{M}{l}\right) dx$$

$$\text{Moment of Inertia of the element about the axis } YY'$$

$$= \left(\frac{M}{l} dx\right) x^2$$

$$\text{Moment of inertia of the whole lamina about the axis } YY'$$

$$I_y = \int_{-l/2}^{+l/2} \frac{M}{l} x^2 dx = \frac{Ml^3}{12} \quad \dots(i)$$

$$\text{Similarly, moment of inertia of the lamina about the axis } XX',$$

$$I_x = \frac{Mb^3}{12} \quad \dots(ii)$$

(b) The moment of inertia of the lamina about an axis (ZZ') passing the centre of gravity and perpendicular to the plane of the lamina, can be calculated by applying perpendicular axes theorem.

$\therefore$

$$I_z = I_x + I_y$$

$$I_z = \frac{M}{12} (l^2 + b^2) \quad \dots(iii)$$

$$I_z = MK^2$$

$$MK^2 = \frac{M}{12} (l^2 + b^2)$$

$$K = \sqrt{\frac{(l^2 + b^2)}{12}} \quad \dots(iv)$$

(c) Moment of inertia of the lamina about an axis AD

$$I_{AD} = I_y + M\left(\frac{l}{2}\right)^2$$

$$I_{AD} = \frac{Ml^3}{12} + \frac{Ml^3}{4} = \frac{Ml^3}{3}$$

(d) Moment of inertia of the lamina about an axis AB

$$I_{AB} = I_x + M\left(\frac{b}{2}\right)^2$$

$$= \frac{Mb^3}{12} + \frac{Mb^3}{4} = \frac{Mb^3}{3}$$

(e) Moment of inertia of the lamina about an axis passing

through the mid-point of one side and perpendicular to plane (Fig. 3.12).

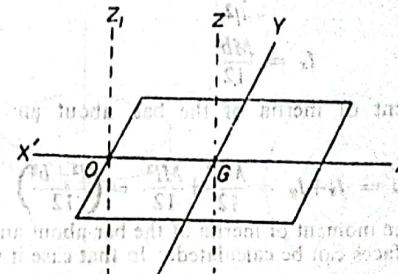


Fig. 3.12

Here

$$I_o = I_z + M\left(\frac{l}{2}\right)^2$$

$$I_o = \frac{M(l^2 + b^2)}{12} + \frac{Ml^2}{4}$$

$$I_o = M\left(\frac{l^2}{3} + \frac{b^2}{12}\right)$$

Note. By using parallel axes theorem, the moment of inertia of the lamina about any axis can be calculated.

3.15 Moment of Inertia of a Solid Uniform Bar of Rectangular cross section

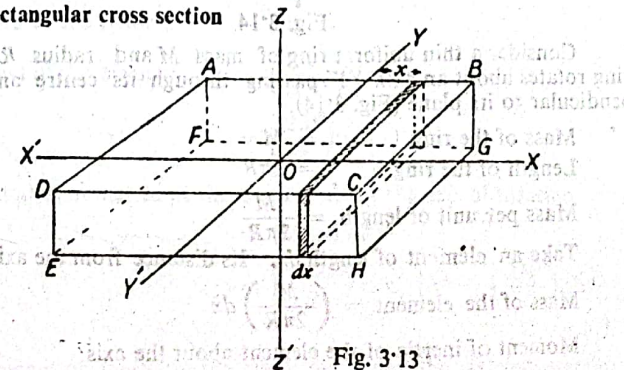


Fig. 3.13

Consider a uniform solid bar of length, l , breadth b and thickness t (Fig. 3.13). Consider an element of length dx , breadth b and thickness t from the axis YY' .

$$\text{Volume of the element} = (dx) b \times t$$

$$\text{Mass of the element} = \left(\frac{M}{lbt}\right) dx \cdot bt = \left(\frac{Mdx}{l}\right)$$

$$I_y = \int_{-b/2}^{b/2} \frac{M}{l} x^2 dx = \frac{Ml^3}{12}$$

$$\text{Similarly } I_z = \frac{Mb^3}{12}$$

$\therefore$ Moment of inertia of the bar about an axis passing through O ,

$$I_x = I_z + I_y = \frac{Mb^3}{12} + \frac{Ml^3}{12} = \frac{M(l^2 + b^2)}{12}$$

Similarly the moment of inertia of the bar about an axis perpendicular to other faces can be calculated. In that case it will be,

$$M\left(\frac{l^2 + b^2}{12}\right) \text{ and } M\left(\frac{b^2 + l^2}{12}\right)$$

3.15. Moment of Inertia of a Ring

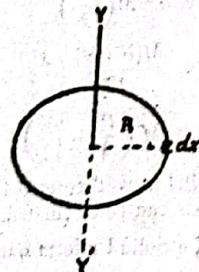


Fig. 3.14

Consider a thin uniform ring of mass M and radius R . The ring rotates about an axis YY' passing through its centre and perpendicular to its plane (Fig. 3.14).

$$\text{Mass of the ring} = M$$

$$\text{Length of the ring} = 2\pi R$$

$$\text{Mass per unit of length} = \frac{M}{2\pi R}$$

Take an element of length dx . Its distance from the axis is R ,

$$\text{Mass of the element} = \left(\frac{M}{2\pi R}\right) dx$$

Moment of inertia of the element about the axis

$$= \left(\frac{M}{2\pi R}\right) dx \cdot R^2$$

Moment of inertia of the ring,

$$I = \int_0^{2\pi R} \frac{MR^2}{2\pi R} dx = MR^2$$

Note. Moment of inertia of a ring about its diameter
 $= \frac{1}{2} MR^2$

3.17 Moment of Inertia of a Uniform circular disc

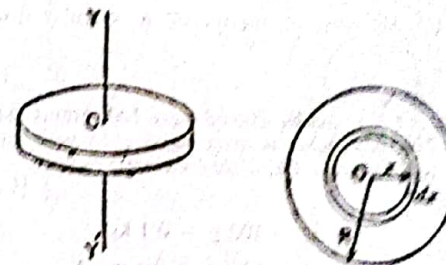


Fig 3.15

Consider a uniform circular disc of mass M and radius R rotating about an axis passing through its centre and perpendicular to its plane (Fig. 3.15).

$$\text{Mass of the disc} = M$$

$$\text{Area of disc} = \pi R^2$$

$$\text{Mass per unit area} = \left(\frac{M}{\pi R^2}\right) \quad \dots (i)$$

Consider a thin element of the disc of radius x and radial thickness dx .

$$\text{Area of the element} = 2\pi x dx$$

$$\text{Mass of the element} = \left(\frac{M}{\pi R^2}\right) 2\pi x dx$$

$$= \left(\frac{2M}{R^2}\right) x dx$$

Moment of inertia of the element about the axis of rotation

$$= (\text{Mass}) (x^2)$$

$$= \left[\left(\frac{2M}{R^2}\right) x dx\right] x^2 = \left(\frac{2M}{R^2}\right) x^3 dx \quad \dots (ii)$$

Moment of inertia of the whole disc about the axis of rotation

$$I = \int_0^R \left(\frac{2M}{R^2}\right) x^3 dx \quad \dots (iii)$$

$$I = \frac{2M}{R^2} \left[\frac{x^4}{4}\right]_0^R = \frac{2M}{R^2} \left[\frac{R^4}{4}\right]$$

$$I = \frac{2M}{R^2} \left[\frac{R^4}{4} \right] = \frac{MR^2}{2}$$

Note. (i) Moment of inertia of a circular disc about its diameter $\frac{MR^2}{4}$

Example 3.2 A circular disc of mass 100 grams and radius 10 cm is making 120 rpm about an axis passing through its centre and perpendicular to its plane. Calculate its kinetic energy.

[Calcutta 1980]

Here,

$$M = 100 \text{ g} = 0.1 \text{ kg}$$

$$R = 10 \text{ cm} = 0.1 \text{ m}$$

$$\text{Number of rotations per second} = \frac{120}{60} = 2$$

$$\omega = 2\pi n = 4\pi \text{ radians/s}$$

$$I = \frac{MR^2}{2} = \frac{0.1 \times (0.1)^2}{2} = 5 \times 10^{-4}$$

$$\text{Kinetic energy} = \frac{1}{2} I \omega^2 = \frac{1}{2} \times 5 \times 10^{-4} \times (4\pi)^2$$

$$= 3.949 \times 10^{-2} \text{ J}$$

Example 3.3 A thin metal ring (hoop) of diameter 0.6 m and mass 1 kg starts from rest and rolls down an inclined plane. Its linear velocity on reaching the foot of the plane is 5 m/s. Calculate (i) the moment of inertia of the ring and (ii) the kinetic energy of rotation at that instant.

Here,

$$M = 1 \text{ kg}$$

$$R = \frac{0.6}{2} = 0.3 \text{ m}, v = 5 \text{ m/s}$$

(i)

$$I = MR^2$$

$$I = 1 \times (0.3)^2$$

$$I = 0.09 \text{ kg-m}^2$$

(ii) Kinetic energy

$$= \frac{1}{2} I \omega^2$$

But

$$\omega = \frac{v}{R}$$

$$\therefore KE = \frac{1}{2} I \left(\frac{v}{R} \right)^2$$

$$= \frac{I v^2}{2 R^2} = \frac{0.09 \times 25}{2 \times (0.3)^2}$$

$$= 12.5 \text{ joules.}$$

(a) About a diameter

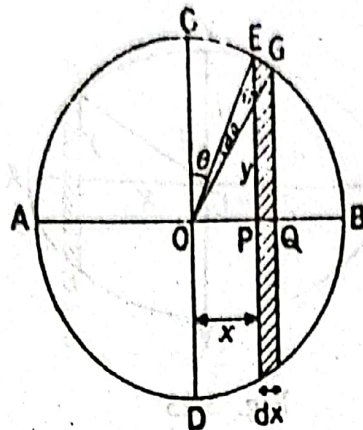


Fig. 3-22!

The radius of the element

$$y = R \cos \theta$$

Also $x = R \sin \theta \quad \therefore dx = R \cos \theta d\theta$

$$EG = R.d\theta$$

Surface area of the element = $2\pi y (EG)$

$$= 2\pi R \cos \theta \cdot R d\theta$$

$$= 2\pi R^2 \cos \theta d\theta = 2\pi R dx$$

$$\text{Mass per unit area of the shell} = \frac{M}{4\pi R^2}$$

$$\text{Mass of the element} = \frac{M}{4\pi R^2} \times 2\pi R dx = \frac{M dx}{2R}$$

Moment of Inertia of the element about the diameter AB

$$= \left(\frac{M dx}{2R} \right) y^2 = \frac{M dx}{2R} (R^2 - x^2)$$

M.I. of the whole shell about the diameter AB,

$$I = 2 \int_0^R \frac{M}{2R} (R^2 - x^2) dx$$

$$I = \frac{M}{R} \left(R^3 - \frac{R^3}{3} \right) = \frac{2}{3} MR^2 \quad \dots(i)$$

(b) About the tangent

$$I_t = I + MR^2$$

$$I_c = \frac{2}{3}MR^2 + MR^2 = \frac{5}{3}MR^2 \quad \dots(ii)$$

$$I = \frac{M \left[\left(\frac{ghr^2}{8\pi^2 n^2} \right) - r^2 \right]}{\left(1 + \frac{n_1}{n} \right)}$$

or

From equation (r), I can be calculated.

3.30 Uses of Flywheel

A flywheel is a heavy wheel capable of rotation about a horizontal or vertical axis. The mass of the wheel is usually concentrated at the rim so as to increase its moment of inertia. The flywheels are provided for stationary engines so as to provide uniform motion. In a four stroke engine, the driving force is provided only during the working stroke. During the rest of the strokes the energy to the driving shaft is provided by the high kinetic energy of the flywheel. The flywheel gains kinetic energy during the working stroke.

Further, in a steam engine, twice in each rotation of the shaft, the force acts at right angles to the shaft. Theoretically, a force does not produce any effect when it acts at right angles to the shaft. For this purpose, a flywheel is used, so that the momentum gained by the flywheel enables the shaft to rotate beyond the dead centres in each half rotation. Thus, the flywheel makes the motion of the shaft smooth and less jerky.

Example 3.5 A flat circular disc of mass 0.05 kg and diameter 0.04 m rolls on its edge on a smooth horizontal surface with a velocity of 0.05 m/s. Calculate its total kinetic energy.

(Calcutta, 1975)

Here

$$M = 0.05 \text{ kg}$$

$$r = 0.01 \text{ m}$$

$$v = 0.05 \text{ m/s}$$

$$\omega = \frac{v}{r} = \frac{0.05}{0.01}$$

$$= 5 \text{ radians/s}$$

$$I = \frac{Mr^2}{2} = \frac{0.05 \times (0.01)^2}{2}$$

$$I = 2.5 \times 10^{-6} \text{ kg-m}^2$$

Total kinetic energy

$$= \frac{1}{2} I \omega^2 + \frac{1}{2} mv^2$$

$$= \frac{1}{2} \times 2.5 \times 10^{-6} \times (5)^2 + \frac{1}{2} \times 0.05 \times (0.05)^2$$

$$= 9.375 \times 10^{-5} \text{ J}$$

Example 3.6 A circular disc of radius 0.06 m can rotate freely about a horizontal axis passing through its centre. A thread

Dynamics of Rigid Bodies

of negligible mass is wound over its circumference and a mass of 0.3 kg is tied to the free end of the thread and allowed to fall freely. In the first 5 seconds, the mass falls through 1 m. Calculate the angular acceleration and moment of inertia of the disc.

(Gorakhpur, 1971)

(i) Here

$$m = 0.3 \text{ kg}$$

$$r = 0.06 \text{ m}$$

$$h = 1 \text{ m}$$

$$t = 5 \text{ s}$$

$$h = \frac{1}{2} at^2$$

$$1 = \frac{1}{2} a (5)^2$$

$$a = 0.08 \text{ m/s}^2$$

$$v = at = 0.08 \times 5 = 0.4 \text{ m/s}$$

$$\omega_1 = 0, \omega_2 = \frac{v}{r} = \frac{0.4}{0.06} = \frac{20}{3} \text{ radians/s}$$

$$a = \frac{\omega_2 - \omega_1}{t} = \frac{20}{3 \times 5} = 1.33 \text{ radian/s}^2$$

(ii)

$$mgh = \frac{1}{2} mv^2 + \frac{1}{2} I \omega^2$$

$$\frac{1}{2} I \omega^2 = mgh - \frac{1}{2} mv^2$$

$$I = \frac{2mgh - mv^2}{\omega^2} = \frac{m[2gh - v^2]}{\omega^2}$$

$$I = \frac{0.3[2 \times 9.8 \times 1 - (0.4)^2]}{(20/3)^2} = \frac{0.3 \times 19.44 \times 9}{400}$$

$$I = 0.1312 \text{ kg-m}^2$$

Example 3.7. A diatomic molecule consists of two atoms of masses m_1 and m_2 kg. The two atoms are at a constant distance of y metres. Calculate the moment of inertia of the system about an axis passing through the centre of gravity of the system and perpendicular to the line joining the atoms.

(Gorakhpur, 1972)

Let the centres of the atoms be at distance a_1 and a_2 from the centre of gravity of the system.

$\therefore$

$$m_1 a_1 = m_2 a_2, \text{ But } a_1 + a_2 = y$$

$\therefore$

$$m_1 a_1 = m_2 (y - a_1)$$

or

$$a_1 = \left(\frac{m_2 y}{m_1 + m_2} \right)$$

Moment of inertia of the system,

$$I = m_1 a_1^2 + m_2 a_2^2 = m_1 a_1^2 + m_2 (y - a_1)^2$$

$$I = m_1 \left(\frac{m_2 y}{m_1 + m_2} \right)^2 + m_2 \left(y - \frac{m_2 y}{m_1 + m_2} \right)^2$$

$$I = m_1 \left(\frac{m_2 y}{m_1 + m_2} \right)^2 + m_2 y^2 + m_2 \left(\frac{m_2 y}{m_1 + m_2} \right)^2 - \frac{2m_1 m_2 y^2}{m_1 + m_2}$$

$$I = \frac{m_1^2 y^2}{m_1 + m_2} + m_2 y^2 - \frac{2m_1 m_2 y^2}{m_1 + m_2} = m_2 y^2 - \frac{m_1^2 y^2}{m_1 + m_2}$$

$$I = m_2 y^2 \left[1 - \left(\frac{1}{m_1 + m_2} \right) \right]$$

Example 3.8. Four spheres each of mass m and radius a are so placed that the centres of the spheres are located at the corners of a square of side b . Calculate the moment of inertia of the whole system about one side taken as the axis. (Gorakhpur, 1977)

Here

$$I = 2 \left(\frac{2}{5} m a^2 \right) + 2 \left[\frac{2}{5} m a^2 + m b^2 \right]$$

$$I = 1.6 m a^2 + 2 m b^2$$

$$I = m [1.6 a^2 + 2 b^2]$$

Example 3.9. A flywheel has a moment of inertia of 1 kg-m^2 . It is rotating at a speed of 2 revolutions/s. Find the constant torque required to stop the wheel in 5 rotations. Calculate also the work done by the braking torque.

Here,

$$I = 1 \text{ kg-m}^2$$

$$n = 2$$

Let the torque be C

$$\text{Work done. } W = C\theta = \frac{1}{2} I \omega^2$$

$$C = \frac{I \omega^2}{2\theta}$$

$$\omega = 2\pi n = 4\pi \text{ radians/s}$$

$$\theta = 5 \times 2\pi = 10\pi \text{ radians}$$

$$C = \frac{1 \times (4\pi)^2}{2 \times 10\pi} = 2.5136 \text{ N-m}$$

$$\text{Work done} = C\theta = 2.5136 \times 10\pi$$

$$= 78.96 \text{ J}$$

Example 3.10. A circular disc of mass M and radius R is set rolling on a table with a velocity v . Show that its total kinetic energy is $0.75 Mv^2$.

$$I = \frac{MR^2}{2}$$

$$\text{Total kinetic energy} = \frac{1}{2} Mv^2 + \frac{1}{2} I \omega^2$$

$$E = \frac{1}{2} Mv^2 + \frac{1}{2} \left(\frac{MR^2}{2} \right) \omega^2$$

$$E = \frac{1}{2} Mv^2 + \frac{1}{4} MR^2 \left(\frac{v^2}{R^2} \right)$$

$$E = 0.75 Mv^2$$

Example 3.11. A solid sphere of mass 1 kg and radius 0.25 m rolls without slipping with a uniform velocity of 0.1 m/s along a straight line on a horizontal table. Calculate its kinetic energy.

Here

$$M = 1 \text{ kg}$$

$$R = 0.25 \text{ m}$$

$$v = 0.1 \text{ m/s}$$

$$I = \frac{2}{5} MR^2$$

Kinetic energy,

$$E = \frac{1}{2} Mv^2 + \frac{1}{2} I \omega^2 = \frac{1}{2} Mv^2 + \frac{1}{2} \left(\frac{2}{5} MR^2 \right) \omega^2$$

$$E = \frac{1}{2} Mv^2 + \frac{1}{2} \times \frac{2}{5} Mv^2$$

$$E = \frac{1}{2} \times \frac{7}{5} Mv^2 = 0.7 Mv^2$$

$$E = 0.7 \times 1 \times (0.1)^2$$

$$E = 7 \times 10^{-3} \text{ J}$$

Example 3.12. A sphere, a disc and a ring of the same radius are allowed to roll down on inclined plane simultaneously from the same height without slipping. Give the order in which they reach the bottom of the plane.

For a body rolling down an inclined plane the acceleration,

$$a = \frac{g \sin \theta}{1 + \frac{K^2}{R^2}}$$

Here K is the radius of gyration.

(i) For a sphere

$$K^2 = \frac{2}{5} R^2 \text{ or } \frac{K^2}{R^2} = \frac{2}{5} = 0.4$$

$$a_1 = \frac{g \sin \theta}{(1+0.4)} = \frac{g \sin \theta}{1.4} \quad \dots(i)$$

(ii) For a disc

$$K^2 = \frac{R^2}{2}$$

$$\frac{K^2}{R^2} = 0.5$$

$$a_2 = \frac{g \sin \theta}{(1+0.5)} = \frac{g \sin \theta}{1.5} \quad \dots(ii)$$

(iii) For a ring

$$K^2 = R^2$$

$$\frac{K^2}{R^2} = 1$$

$$a_3 = \frac{g \sin \theta}{1+1} = \frac{g \sin \theta}{2} \quad \dots(iii)$$

From equation (i), (ii) and (iii)

$$a_1 > a_2 > a_3$$

Therefore, the order of their reaching the bottom of the plane is sphere, disc and ring.

Example 3.13. A solid sphere and a hollow sphere roll down an inclined plane from the same height. Calculate the ratio of their accelerations.For a solid sphere, $K_1^2 = \frac{1}{2} R_1^2$

$$a_1 = \frac{g \sin \theta}{\left(1 + \frac{K_1^2}{R_1^2}\right)} = \frac{g \sin \theta}{1+0.4}$$

$$a_1 = \frac{g \sin \theta}{1.4} \quad \dots(i)$$

For a hollow sphere, $K_2^2 = \frac{2}{3} R_2^2$

$$a_2 = \frac{g \sin \theta}{\left(1 + \frac{K_2^2}{R_2^2}\right)} = \frac{g \sin \theta}{1+0.67} = \frac{g \sin \theta}{1.67} \quad \dots(ii)$$

Dividing (i) by (ii)

$$\frac{a_1}{a_2} = \frac{1.67}{1.4}$$

$$\frac{a_1}{a_2} = 1.19$$

Example 3.14 A solid sphere rolls down an inclined plane with an acceleration of 3.5 m/s^2 . Calculate the inclination of the plane. Also calculate the acceleration of a hollow sphere rolling down the same inclined plane.

For a solid sphere rolling down an inclined plane

$$a = \frac{g \sin \theta}{\left(1 + \frac{K^2}{R^2}\right)}$$

$$a = \frac{g \sin \theta}{(1+0.4)} = \frac{g \sin \theta}{1.4}$$

$$3.5 = \frac{9.8 \sin \theta}{1.4}$$

$$\sin \theta = 0.5$$

$$\theta = 30^\circ$$

For a hollow sphere

$$a = \frac{g \sin \theta}{\left(1 + \frac{K^2}{R^2}\right)} = \frac{g \sin \theta}{(1+0.67)} = \frac{g \sin \theta}{1.67}$$

$$a = \frac{9.8 \times 0.5}{1.67}$$

$$a = 2.94 \text{ m/s}^2$$

Example 3.15 A thin hollow cylinder open at both ends and weighing 10 kg (a) slides with a speed of 10 m/s without rotating (b) rolls with a speed of 10 m/s without slipping. Calculate the kinetic energy of the cylinder in each case.

Here

$$M = 10 \text{ kg}$$

$$v = 10 \text{ m/s}$$

(a) Kinetic energy,

$$E = \frac{1}{2} Mv^2 = \frac{1}{2} \times 10 (10)^2$$

$$E = 500 \text{ J}$$

(b) Kinetic energy

$$E = \frac{1}{2} Mv^2 + \frac{1}{2} I\omega^2$$

In the case of a hollow cylinder $I = MR^2$

$$\therefore E = \frac{1}{2} Mv^2 + \frac{1}{2} MR^2 \omega^2 = \frac{1}{2} Mv^2 + \frac{1}{2} Mv^2$$

$$E = Mv^2 = 10 \times (10)^2$$

$$E = 1000 \text{ J}$$

Example 3.16 A flywheel of mass 100 kg and radius 0.5 m makes 10 revolutions per second. Calculate (i) angular velocity.

(ii) moment of inertia and (iii) the energy of the flywheel, that the whole mass of the flywheel is concentrated at the rim. Assume

Here

$$M = 100 \text{ kg}$$

$$R = 0.5 \text{ m}$$

$$n = 10$$

(i) Angular velocity

$$\omega = 2\pi n = 2\pi \times 10 = 20\pi$$

$$\omega = 62.84 \text{ rad/s}$$

(ii) Moment of inertia

$$I = MR^2 = 100 \times (0.5)^2$$

$$I = 25 \text{ kg-m}^2$$

(iii) Energy

$$E = \frac{1}{2} I \omega^2$$

$$E = \frac{1}{2} \times 25 (62.84)^2$$

$$E = 4.938 \times 10^4 \text{ J}$$

Example 3.17. A flywheel is accelerated by a steady torque of 25 N-m so that it makes 3 revolutions in the first second of its motion. The mass of the flywheel is 15 kg. Calculate the radius of gyration of the flywheel.

Here

$$\theta = 6\pi$$

$$t = 1 \text{ second}; \quad \theta = \frac{1}{2} a t^2$$

$$a = \frac{2\theta}{t^2} = \frac{2 \times 6\pi}{1}$$

$$a = 12\pi \text{ radians/s}^2$$

$$\text{Torque} = I a = 25 \text{ N-m}$$

$$\therefore I = \frac{25}{12\pi} \text{ kg-m}^2$$

But

$$I = MK^2$$

$$K^2 = \frac{I}{M}$$

$$K = \sqrt{\frac{I}{M}} = \sqrt{\frac{25}{12\pi \times 15}}$$

$$K = 0.2102 \text{ m}$$

Example 3.18. An aircraft propeller has a mass of 80 kg and radius of gyration 0.75 m. Calculate its moment of inertia. Calculate also the magnitude of an unbalanced torque that can produce an angular acceleration of 40 radians/s².

Here

$$I = MK^2$$

$$M = 80 \text{ kg}$$

$$K = 0.75 \text{ m}$$

$$\begin{aligned}
 \therefore \text{Torque} \quad I &= 80 \times (0.75)^2 = 45 \text{ kg-m}^2 \\
 \tau &= I\alpha \\
 \alpha &= 40 \text{ radians/s}^2 \\
 \therefore \tau &= 45 \times 40 \\
 &= 1800 \text{ N-m}
 \end{aligned}$$

Example 3.19. A wheel and axle is made to rotate about a horizontal axis with the help of a body of mass 5 kg attached to a string wound around the axle. The radius of the axle is 10 cm. The body falls vertically through 5 metres in 10 seconds starting from rest. Calculate the moment of inertia of the wheel and axle.

Linear acceleration of the body = a

$$\begin{aligned}
 h &= \frac{1}{2}at^2 \\
 \text{Here } h &= 5 \text{ m, } t = 10 \text{ s,} \\
 \therefore a &= \frac{2h}{t^2} = \frac{2 \times 5}{100} \\
 a &= 0.1 \text{ m/s}^2
 \end{aligned}$$

Angular acceleration.

$$a = \frac{a}{r}$$

Here $a = 0.1 \text{ m/s}^2$, $r = 10 \text{ cm} = 0.1 \text{ m}$

$$\begin{aligned}
 \therefore a &= \frac{0.1}{0.1} \\
 a &= 1 \text{ radian/s}^2
 \end{aligned}$$

Also $Mg - T = Ma$

$$T = Mg - Ma = M(g - a)$$

Here $M = 5 \text{ kg}$

$$T = 5(9.8 - 0.1)$$

$$T = 48.5 \text{ N}$$

Torque

$$= T \times r = I\alpha$$

$$I = \frac{Tr}{\alpha} = \frac{48.5 \times 0.1}{1}$$

$$I = 4.85 \text{ kg-m}^2$$

Example 3.20. The stiffness of a shaft is such that a torque of 1000 N-m is required for 1 radian twist. Calculate the energy required to twist the shaft through an angle of 36° assuming that the twisting moment is proportional to the angle of twist.

Torque per radian = 10^3 N-m/radian

$$\theta = 36^\circ = \frac{36 \times \pi}{180} = \frac{\pi}{5} \text{ radian}$$

Maximum torque acting on the shaft

$$= \frac{10^3 \times \pi}{5} = 200\pi \text{ N-m}$$

As the angle of twist increases, the torque increases proportionately. The average torque = $\frac{1}{2} \times$ maximum torque

$$\therefore \text{Average torque} = \frac{200\pi}{2} = 100\pi \text{ N-m}$$

$$\text{Energy required} = \text{Work done} \\ = \text{Average torque} \times \text{angle}$$

$$= 100\pi \times \frac{\pi}{5} = 20\pi^2$$

$$= 197.55 \text{ J}$$

Example 3.21. The driving wheel of a belt attached to an electric motor has a radius of 0.25 m. It makes 40 revolutions per second. The tension in the belt is 200 N on slack side and 1000 N on tight side. Calculate the power transmitted by the belt.

$$F = 1000 - 200 = 800 \text{ N}$$

$$r = 0.25 \text{ m}$$

$$\text{Torque, } \tau = F \times r = 800 \times 0.25$$

$$= 200 \text{ N-m}$$

$$\omega = 40 \text{ revolutions/s} = 40 \times 2\pi$$

$$= 80\pi \text{ radians/s}$$

$$\text{Power, } P = \tau \times \omega$$

$$P = 200 \times 80\pi \text{ watts}$$

$$P = 50.24 \text{ kW}$$

Example 3.22. Calculate the power transmitted by a rope that is being wound on a wheel of circumference 5 metres when the wheel makes 2 revolutions per second. Tension in the rope is 50 N.

$$\text{Distance moved in one second, } s = 2 \times 5 = 10 \text{ m/s}$$

$$T = 50 \text{ N}$$

$$\text{Power} = T \times s = 50 \times 10$$

$$= 500 \text{ watts}$$

Exercises 3

1. What is a rigid body?
2. Distinguish between mass of a body and moment of inertia of a body.
3. Explain the terms moment of inertia and radius of gyration. Discuss the physical significance of moment of inertia.
4. State and explain the law of conservation of angular momentum. Give examples.

[Delhi 1980]

5. State and prove the theorem of perpendicular axis in moment of inertia.
6. State and prove the theorem of parallel axis in moment of inertia. [Mysore, 1978]
7. Prove that, out of an infinite number axes which can be imagined parallel to a given direction, the moment of inertia of a body is least. About the one passing through its centre of gravity. [Delhi Engineering, 1979]
8. Define moment of inertia and radius of gyration of a rotating body. There are two parallel axes. The first one passes through C.G. of the body about which axis the body will have larger moment of inertia. [Calcutta, 1980]
9. Define moment of inertia and radius of gyration. State and prove theorems of perpendicular and parallel axes. [Kanpur, 1981, Delhi (Sub) 1980]
10. Derive an expression for the moment of inertia of a thin uniform bar about an axis passing through one end and perpendicular to its length.
11. Derive an expression for the moment of inertia of a rectangular lamina about an axis passing through its centre and perpendicular to its plane.
12. Show that the moment of inertia of a uniform circular disc about an axis passing through its centre and perpendicular to its plane is $0.5 MR^2$.
13. Derive an expression for the moment of inertia of a solid cylinder about an axis passing through its centre and perpendicular to its length.
14. Define (a) angular momentum (b) rotational kinetic energy. Find the moment of inertia of a solid sphere about its diameter. [Calcutta, 1978]
15. A solid sphere rolls on a horizontal plane without slipping. Calculate the ratio of its translational kinetic energy to the rotational kinetic energy. [Calcutta, 1978]
16. Calculate the moment of inertia of a circular disc about its diameter. [Calcutta, 1980]
17. Define moment of inertia of a rotating body. What is its physical significance? Calculate the moment of a sphere about (i) its diameter and (ii) a tangent. [Kanpur 1981, Delhi (Hons) 1980]
18. Show that the moment of inertia of a sphere about any tangent

$$= \left(\frac{7}{5}\right) MR^2$$

- 94
19. Derive an expression for the moment of inertia of a spherical shell about its diameter.
 20. Show that the moment of inertia of an elliptical lamina about its minor axis $= 0.25 Ma^2$.
 21. Derive an expression for the moment of inertia of a triangular lamina about an axis coincident with one side.
 22. Show that the moment of inertia of a solid cone about an axis passing through its vertex and perpendicular to the base is $= \frac{3MR^2}{10}$.
 23. Calculate the total kinetic energy of a circular disc rolling down an inclined plane.
 24. A solid cylinder of mass M rolls without slipping down a plane of length L inclined at an angle θ with the horizontal. What is the velocity of the centre of mass of the cylinder at the bottom? [Delhi (Hons) 1980]
 25. Discuss a method to determine the moment of inertia of a flywheel. What are the practical uses of a flywheel?
 26. Explain the term moment of inertia of a rotating body and give two illustrations to show that this term is analogous to mass in the case of linear motion. [Delhi 1976]
 27. Define moment of inertia and radius of gyration. State and prove the theorem of parallel axis as applied to moment of inertia. [Delhi (sub) 1976, 78, 80]
 28. For each of following pairs of the bodies, state which has the larger moment of inertia about the axis of symmetry passing through the centre of the body. Give an explanation also.
 - (a) Two cylinders with the same mass, the same outer radius and the same length but one of the cylinders is hollow.
 - (b) Two homogeneous cylinder of the mass and the same outer radius but of different lengths. [Delhi (Sub) 1977]
 29. Calculate the moment of inertia of a uniform thin rigid bar of mass M and length l , about an axis passing through the centre of gravity of the bar and perpendicular to its length. Show that if equal point masses are attached at the ends of the ends of the rod, such that the mass of the loaded bar becomes $\frac{4M}{3}$, the moment of inertia about the same axis is doubled. (Delhi 1979)
 30. A thin uniform circular ring having moment of inertia $\frac{30}{\pi^2}$ kg-m² about an axis passing through the centre and perpendicular

Dynamics of Rigid Bodies

95

- to the plane of the ring is straightened to form a bar. Calculate its moment of inertia passing through the centre of gravity of the bar and perpendicular to its length. (Delhi 1979) [Ans. 10 kg-m²]
31. Derive an expression for the moment of inertia of a sphere about its diameter. [Delhi (Hons) 1979]
32. A uniform sphere of mass M and radius R and uniform circular disc of mass M and radius R are released simultaneously from rest at the top of an inclined plane. Which one will reach the bottom first if both roll without slipping? [Delhi (Hons) 79] [Ans. Sphere reaches the bottom first]
33. Calculate the moment of inertia of a thin uniform circular disc of mass 5 kg and radius 0.1 m rotating about an axis passing through its centre and perpendicular to its plane. [Ans. 0.025 kg-m²]
34. A thin metal ring has a mass of 2 kg and diameter 0.8 m. The ring starts from rest and rolls down an inclined plane. The linear velocity on reaching the foot of the plane is 10 m/s. Calculate (i) the moment of inertia of the ring and (ii) kinetic energy of rotation at the foot of the plane. [Ans. (i) 0.32 kg-m² (ii) 100 J]
35. A flywheel of an engine starts from rest and acquires an angular velocity of 54 radians/s in 5 seconds. Calculate (i) angular acceleration and (ii) angular displacement in 5 seconds [Ans (i) 10.8 radians/s² (ii) 125 radians]
36. A uniform circular disc of mass 10 kg and radius 0.2 m is subjected, to a torque consisting of 2 equal and opposite forces acting tangentially to the circumference of the disc at two opposite ends of a diameter. The disc can freely rotate about an axis passing through its centre and perpendicular to its plane. Calculate (i) the magnitude of each force required so that the disc can rotate with an angular acceleration of 20 radians/s² and (ii) the kinetic energy of the disc after 10 seconds from the start. [Ans. (i) 10 N (ii) 4000 J]
37. A thin uniform circular disc of mass 10 kg and radius 0.1 m rotates about an axis passing through its centre and perpendicular to its plane. Calculate its moment of inertia. [Ans. 0.05 kg-m²]
38. A thin metal hoop (ring) of diameter 0.8 m and mass 1 kg starts from rest and rolls down an inclined plane. Its linear velocity on reaching the foot of the ring and (ii) the kinetic energy of rotation at that instant. [Ans. (i) 0.16 kg-m² (ii) 50 J]

39. A flywheel of an engine starts from rest and attains an angular velocity of 200 radians/s in 5 s. Calculate (i) the angular acceleration and (ii) the angular displacement in 5 seconds.

[Ans. (i) 40 radians/s²
(ii) 500 radians]

40. A thin uniform circular disc of mass 10 kg and radius 5 cm is subjected to a torque. The torque is due to two forces acting tangentially to the circumference of the disc at the two diametrically opposite ends. The disc can freely rotate about an axis passing through its centre and perpendicular to its plane. Calculate the magnitude of each force required so that the disc can rotate with angular acceleration of 10 radians/s². Calculate also the kinetic energy of rotation the rotation of the disc after 5 seconds from the start.

[Ans. (i) 1.25 N
(ii) 15.625 J]

41. A grinder is in the form of a circular disc of mass 5 kg and diameter 0.5 m. Calculate the constant torque that has to be applied so that the disc acquires an angular velocity of 5 revolutions/s in 10 seconds. Calculate also the rate at which work is done by the torque at the end of 10 seconds.

[Ans. (i) 0.4742 N-m
(ii) 1.489 W]

42. A flywheel has a mass 10 kg and diameter 0.2 m. The flywheel is making 300 revolutions per minute (rpm). Calculate (i) the torque that is required to acquire this speed in 10 seconds and (ii) the force applied tangentially at the rim of the flywheel.

$$\left[\begin{array}{l} \text{Hint : } I = \frac{MR^2}{2}, \alpha = \frac{\omega_1 - \omega_2}{t} \\ \text{Torque} = I\alpha = F \times R \end{array} \right]$$

[Ans. (i) 0.1571 N-m
(ii) 1.571 N]

43. A flat circular disc of mass 0.1 kg and diameter 0.1 m rolls on its edge on a smooth horizontal surface with a velocity of 5 m/s. Calculate its total kinetic energy.

[Ans. 1.875 J]

44. A circular disc of radius 0.08 m can rotate freely about a horizontal axis passing through its centre. A thread of negligible mass is wound over its circumference and a mass of 0.25 kg is tied to the free end of the thread and allowed to fall freely. In the first 4 seconds the mass falls through 2 m. Calculate (i) angular acceleration and (ii) moment of inertia of the disc.

[Ans. (i) 3.125 radian/s²
(ii) 0.0611 kg-m²]

Dynamics of Rigid Bodies

45. A diatomic molecule consists of two atoms of masses m_1 and m_2 kg. The two atoms are at a constant distance of x metres. Calculate the moment of inertia of the system about an axis passing through the centre of gravity of the system and perpendicular to the line joining the atoms.

$$\text{Ans. } I = m_1 x^2 \left[1 - \frac{1}{(m_1 + m_2)} \right]$$

46. Four spheres each of mass m kg and radius b metres are so placed that the centres of the spheres are located at the four corners of a square of side a metres. Calculate the moment of inertia of the whole system about one side of the square taken as the axis.

[Ans. $I = m(1.6b^2 + 2a^2)$]

47. A flywheel has a moment of inertia of 5 kg-m². It is rotating at a speed of 5 revolutions/s. Calculate the constant torque required to stop the flywheel in 10 rotations. Calculate also the work done by the braking torque.

[Ans. (i) 39.17 N-m (ii) 2467.41 J]

48. A circular disc of mass 5 kg and radius 0.05 m is set rolling on a table with a velocity of 5 m/s. Calculate the total kinetic energy of the disc.

[Ans. 93.75 J]

49. A solid sphere of radius 0.1 m and mass 2 kg rolls without slipping with a uniform velocity of 0.2 m/s along a straight line on a horizontal surface. Calculate its kinetic energy.

[Ans. 0.056 J]

50. A solid sphere rolls down an inclined plane with an acceleration of 6.062 m/s².

- (i) Calculate the inclination of the plane.
(ii) Calculate also the acceleration of a hollow sphere rolling down the same inclined plane.

[Ans. (i) 60° (ii) 5.08 m/s²]

51. A thin hollow cylinder open at both ends and weighing 5 kg (a) slides with a speed of 5 m/s without rotating (b) rolls with a speed of 5 m/s without slipping. Calculate the kinetic energy of the cylinder in each case.

[Ans. (i) 62.5 J (ii) 125 J]

52. An aircraft propeller has a mass of 100 kg and radius of gyration 1 m. Calculate its moment of inertia. Calculate also the magnitude of an unbalanced torque that can produce an angular acceleration of 20 radians/s².

[Ans. (i) 100 kg-m² (ii) 2000 N-m]

53. The stiffness of a shaft is such that a torque of 500 N-m is required for 0.5 radian twist. Calculate the energy required to

twist the shaft through an angle of 18° assuming that the twisting moment is proportional to the angle of twist.

[Ans. 49.39 J]

54. The driving wheel of a belt attached to an electric motor has a radius of 0.5 m. It makes 20 revolutions per second. The tension in the belt is 400 N on the slack side and 1200 N on the tight side. Calculate the power transmitted by the belt.

[Ans. 5.0265×10^4 W]

55. Calculate the power transmitted by a rope that is being wound on a wheel of circumference 2.5 m when the wheel makes 4 revolutions per second. Tension in the rope is equal to 100 N.

[Ans. 1000 W]

56. State the law of conservation of angular momentum and illustrate with an example. [Delhi (Hons) 1982]

57. (a) Distinguish between 'mass' and 'moment of inertia' of a body.

- (b) State and prove the theorem of parallel axes for moment of inertia.

- (c) What are the practical uses of a flywheel? [Delhi (Sub) 1982]

58. (a) Define moment of inertia and explain its physical significance. [Meerut, 1981]

- (b) Deduce the moment of inertia of a solid sphere about a tangent drawn on its surface. [Meerut, 1981]

59. A cylinder has mass M , length L and radius R . Calculate the moment of inertia of this cylinder about an axis passing through its centre of mass and perpendicular to its length.

[Lucknow, 1982]

60. Define moment of inertia of a body. State and prove the theorem of parallel axis. [Rohilkhand, 1982]

61. A hollow sphere and a circular hoop of the same mass and radii start rolling down a smooth inclined plane simultaneously. Obtain expression for their acceleration and determine which one will reach the bottom earlier. [Rohilkhand, 1982]

62. State and prove the theorem of parallel axes. About which axis, out of a number of parallel axes, will a body have the least moment of inertia? [Delhi (Hons) 1984]

63. Calculate the moment of inertia of a uniform solid spherical body about an axis passing through its centre.

[Delhi (Hons) 1984]

64. Calculate the moment of inertia of a solid cylinder about an axis passing through one of its ends and perpendicular to its length.

[Delhi (Hons) 1985]